



## Practice assignment 8 - Not Graded



This assignment will not be graded and is only for practice.

Note: Suppose  $W_1$  and  $W_2$  are subspaces of a vector space  $V$ . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that  $W_1 + W_2$  is a subspace of  $V$ .

### Multiple Choice Questions (MCQ):

1 point

An inner product on  $R^3$  is defined as:

$$\langle \cdot, \cdot \rangle: R^3 \times R^3 \rightarrow R$$

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

Match a set  $S$  in Column A with the set  $T$  given in column B, so that  $T \subset \text{Span}(S)^\perp$ . Match a set in column B with the property it satisfies in column C.



Which of the following option are true?

☐

d → ii → 3

☐

d → iii → 4

☐

c → ii → 4

☐

a → iv → 1

☐

a → ii → 3

☐

b → iv → 2

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

d → iii → 4

a → ii → 3

Suppose  $W_1$  and  $W_2$   $W_1$  and  $W_2$  are subspaces of a vector space  $V$ . Let  $P_{W_1}$  and  $P_{W_2}$   $P_{W_1}$  and  $P_{W_2}$  denote the projection from  $V$  to  $W_1$  and  $V$  to  $W_2$  respectively and consider the following statements:

- **Statement P:** If  $P_{W_1} + P_{W_2}$   $P_{W_1} + P_{W_2}$  is a projection from  $V$  to  $W_1 + W_2$ , then  $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$ .  $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$ .
- **Statement Q:**  $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$   $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$  is not a projection from  $V$  to  $W_1$ .

- **Statement R:** If  $AA$  is the matrix representation of  $P_{W_2} P_{W_1}$ , then  $AA$  can not be a symmetric matrix.
- **Statement S:**  $P_{W_1} - P_{W_2} P_{W_1} - P_{W_2}$  is a projection from  $V$  to  $W_1 + W_2$ .

Find the number of correct statements

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Which of the followings option(s) is (are) true?

☐

Let  $u, v$ , and  $w$  be three vectors in  $R^2$  and  $\langle \cdot, \cdot \rangle$  be an inner product on  $R^2$ . If  $\langle u, v \rangle = \langle u, w \rangle$  and  $\langle v, w \rangle = \langle u, w \rangle$ , then  $v = w$

☐

Let  $V$  be an inner product space. If  $\langle u, v \rangle = 0$  for all  $u \in V$ , then  $v = 0$

☐

Let  $V$  be an inner product space and  $u, v \in V$ . If  $\langle u + v, u - v \rangle = 0$ , then  $\|u\| \neq \|v\|$

☐

Let  $V$  be an inner product space. If  $S \subset V$  then  $S^\perp = (\text{span}(S))^\perp$

☐

Suppose  $\{u_1, u_2, \dots, u_m\}$  be an orthogonal set of vectors in an inner product space  $V$ , then  $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Let  $V$  be an inner product space. If  $\langle u, v \rangle = 0$  for all  $u \in V$ , then  $v = 0$

Let  $V$  be an inner product space. If  $S \subset V$  then  $S^\perp = (\text{span}(S))^\perp$

Suppose  $\{u_1, u_2, \dots, u_m\}$  be an orthogonal set of vectors in an inner product space  $V$ , then  $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$

1 point

Which of the following option(s) is/are true?

☐

Let  $A$  and  $B$  be orthogonal matrices of order 3. Then  $A$  and  $BA$  and  $B$  are equivalent matrices.

☐

Let  $A$  and  $B$  be orthogonal matrices. Then  $A$  and  $BA$  and  $B$  are similar matrices.

☐

Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle  $\theta$ . Then nullity of the matrix  $A$  is 0.

☐

Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle  $\theta$  and  $B$  be a rotation matrix corresponding to the anti-clock wise rotation of the YZ-plane about the X-axis with angle  $\beta$ . Then  $A$  and  $B$  are equivalent matrices.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Let  $A$  and  $B$  be orthogonal matrices of order 3. Then  $A$  and  $BA$  and  $B$  are equivalent matrices.

Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the  $XY$ -plane about the  $Z$ -axis with angle  $\theta$ . Then nullity of the matrix  $A$  is 0.

Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the  $XY$ -plane about the  $Z$ -axis with angle  $\theta$  and  $B$  be a rotation matrix corresponding to the anti- clock wise rotation of the  $YZ$ -plane about the  $X$ -axis with angle  $\beta$ . Then  $A$  and  $B$  are equivalent matrices.

**Numerical Answer Type:**

Let  $u = (1, 2, 1)$  and  $v = (1, 2, 3)$  be the vectors of the inner product space  $R^3$  with usual inner product. Let  $T: R^3 \rightarrow R^3$  be an orthogonal linear transformation. If  $\theta$  is an angle between  $T(u)$  and  $T(v)$ , then find the value of  $||T(u)|| ||T(v)|| \cos \theta$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 8

1 point

Consider a set  $W = \{(1, 1, 1)\}$  from the inner product space  $R^3$ . Find the dimension of the subspace  $W^\perp$ , where  $W^\perp$  is the collection of the vectors which are orthogonal to the vector  $(1, 1, 1)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

Let  $v = (3, 1, 2)$  be a vector in  $R^3$ . If  $(a, b, c)$  is the vector obtained from  $v$  after the anti- clock wise rotation of  $ZX$ -plane with angle  $45^\circ$  about the  $Y$ - axis, then find the value of  $\sqrt{2}(a + b + c - 1)$ .

$(a + b + c - 1)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 6

1 point

**Comprehension Type Question:**

With a particular frame of reference (in  $R^2$ ) the position of Rahul's house is at the point  $(8, 1)$ . There are two roads Road 1 and Road 2 along the lines  $y = 2x$  and  $y = -2x$  respectively on the  $XY$ -plane.

Answer questions 8,9 and 10 using the given information.

1 point

Which of the following option(s) is (are) true?

☐ If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

☐

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at  $\frac{6}{5}(1, -2)$   $\frac{6}{5}(1, -2)$ .

☐

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at  $\frac{6}{5}(1, -2)$   $\frac{6}{5}(1, -2)$ .

☐ If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at  $\frac{6}{5}(1, -2)$   $\frac{6}{5}(1, -2)$ .

Let  $\theta$  be the acute angle between Road 1 and Road 2, then what will be the value of  $5\cos(\theta)$   $5\cos(\theta)$ ?

**No, the answer is incorrect.**

**Score: 0**